Modular Anthropogenic Processing Depots

Bringing Scale-Appropriate Reprocessing to the Feedstock

Metatron Dynamics — Investor Proposition Author: Robin Macomber | Date: March 2026

Executive Summary

The conventional model for materials reprocessing assumes centralized facilities requiring high capital investment, large feedstock volumes, and wide-catchment logistics. This model replicates the scale assumptions of geological extraction — and inherits its structural limitations: high capital risk, long deployment timelines, feedstock supply uncertainty, and logistics costs that erode the energy-cost advantage that makes reprocessing profitable in the first place.

This proposition inverts the model.

Instead of moving feedstock to scale, we deploy scale-appropriate processing to where the feedstock already is.

The Modular Anthropogenic Processing Depot (MAPD) is a containerized, configurable system comprising standardized modules that can be combined to match any local feedstock inventory. A small rural landfill deploys two modules. A large urban transfer station deploys ten. The modules are standardized. The configuration is variable. Deployment time is measured in days, not years.

The MAPD applies the Metatron Dynamics framework at every level: the local anthropogenic reservoir is declared as a bounded domain before any processing configuration is selected. This ensures that every deployment operates within its valid operating envelope — matching processing capacity to actual feedstock composition, volume, and local demand.

The result is a reprocessing network that scales by multiplication of bounded units rather than expansion of unbounded facilities — eliminating the single largest barrier to capturing the energy-cost arbitrage described in *The Anthropogenic Reservoir* white paper.

The Problem with Centralized Reprocessing

The energy economics of anthropogenic feedstock are established. Recycled aluminum

requires approximately 5% of primary production energy. Scrap steel via electric arc furnace uses roughly 25–35% of blast furnace energy. Plastic pyrolysis bypasses four of five primary processing steps for hydrocarbon recovery. These are not marginal savings — they are structural advantages that widen as geological extraction becomes more energy-intensive.

Yet the reprocessing industry captures only a fraction of this advantage. The reason is not technological. The barrier is architectural: the dominant model requires building large, fixed facilities and then solving the logistics problem of aggregating sufficient feedstock to keep them running.

This creates three structural failure modes:

Feedstock mismatch. A centralized facility is designed for a specific throughput of specific material types. The actual feedstock arriving from a wide catchment area varies in composition, contamination level, and volume. The facility operates against an undeclared domain — the gap between designed capacity and actual feedstock is discovered operationally, not structurally.

Logistics cost erosion. The energy advantage of reprocessing over extraction is real but finite. Transportation costs — collecting, hauling, and delivering feedstock from distributed sources to a central facility — directly erode that advantage. For lower-value materials (mixed plastics, low-grade scrap), logistics costs can consume the entire margin.

Capital concentration risk. A large centralized facility represents concentrated capital deployment against uncertain feedstock supply. If feedstock composition shifts, if regulatory conditions change, or if local supply contracts fall short, the entire investment is at risk. This is the same risk profile as geological extraction — exploration risk replaced by aggregation risk.

Each of these failure modes has the same structural cause: the processing system is not declared as a bounded domain matched to a characterized feedstock inventory. The system is built first and the domain is discovered later.

The MAPD Architecture

Design Principle

> Declare the domain before deploying the processing.

Every MAPD deployment begins with characterization of the local anthropogenic reservoir — composition, volume, contamination profile, seasonal variation. This characterization is

Operator A: explicit domain declaration. The processing configuration is then selected to match that declared domain, not the other way around.

Module Types

The MAPD system comprises five standardized module types, each housed in standard shipping containers (20ft or 40ft) for transport by truck, rail, or ship.

Module 1 — Characterization and Sorting

This module deploys first in every configuration. It is Operator A applied to the physical feedstock — declaring what is actually present before any processing begins.

Technology: NIR (near-infrared) optical sorting with AI-assisted material identification, density separation, magnetic separation for ferrous metals, eddy current separation for non-ferrous metals. Current commercial systems achieve purity rates above 95% and process throughputs compatible with container-scale operations.

Function: Receives mixed waste or landfill-extracted material and produces characterized, sorted feedstock streams — separated polyolefins (PE, PP, PS), excluded contaminants (PVC, PET routed to appropriate handling), ferrous metals, non-ferrous metals (aluminum, copper), and residual fraction.

Output: Declared feedstock streams with known composition, ready for material-specific processing modules.

Why it deploys first: You cannot process what you have not declared. A processing module operating on uncharacterized feedstock is B without A — locally correct operations on an undeclared domain. The sorting module ensures every downstream operation runs within a bounded, known input domain.

Module 2 — Plastics Pyrolysis

Technology: Continuous-feed containerized pyrolysis reactor operating at 400–600°C in an oxygen-free environment. Commercial containerized units exist at scales from 1 ton/day to 10+ tons/day per container. Energy input for thermal decomposition is approximately 1.3–1.8 MJ per kg of plastic feedstock (0.37–0.5 kWh/kg), representing less than 3% of the calorific value of the resulting products. Syngas byproduct can be recirculated to partially sustain the thermal process, and optimized configurations approach thermal self-sufficiency.

Input: Sorted polyolefins (PE, PP, PS) from Module 1.

Output: Pyrolysis oil (38–43 MJ/kg heating value, comparable to petroleum diesel), syngas (combustible, recyclable to process heat), and carbon black (solid byproduct with industrial market value).

Conversion: 75–85% of suitable plastic feedstock converts to liquid oil at current commercial yields.

Module 3 — Metals Reprocessing

Technology: Induction furnace melting — electromagnetic heating that generates heat directly within the metal without flame contact. Commercially available in capacities from 50 kg to 500+ kg per batch, with containerizable footprints. Aluminum melting at 660°C requires approximately 5% of primary production energy. Copper remelting requires approximately 10–15% of primary production energy. Induction furnaces achieve over 90% energy efficiency, produce minimal emissions (0.8 kg CO_2 per ton of clean scrap), and provide precise temperature control for alloy-quality output.

Input: Sorted ferrous and non-ferrous metals from Module 1.

Output: Metal ingots or billets of known composition, suitable for direct industrial use or sale to secondary metals markets.

Recovery rates: 92–95% metal recovery from clean sorted scrap via induction melting.

Module 4 — Electronics and High-Value Recovery *(Optional — deployed where e-waste concentration warrants)*

Technology: Mechanical size reduction, magnetic and eddy current separation, and hydrometallurgical recovery for precious metals and rare earth elements. E-waste contains concentrations of gold, palladium, platinum, and rare earths that exceed natural ore grades by factors of 10–100x.

Input: Electronics waste separated by Module 1.

Output: Recovered precious metals, copper, and rare earth concentrates. Higher-value output per ton than any other module.

Deployment trigger: Local e-waste volume sufficient to justify module capital cost — typically urban or peri-urban locations with electronics discard streams.

Module 5 — Utilities and Output Handling

Technology: Power distribution, cooling systems, output product storage and handling, environmental monitoring and emissions control.

Function: Shared infrastructure serving all processing modules. Syngas from Module 2 can be routed to Module 5 for power generation, reducing or eliminating external energy dependency for the depot.

Environmental compliance: Integrated emissions monitoring and scrubbing. Containerized format allows standardized compliance engineering across deployments.

Configuration Logic

The MAPD system scales by configuration, not by facility expansion.

Deployment Context	Typical Configuration	Modules	Estimated Throughput
Small rural landfill	Sorting + Pyrolysis	2–3 containers	1–3 tons/day
Municipal transfer station	Sorting + Pyrolysis + Metals	4–6 containers	5–15 tons/day
Large urban site	Full suite including e-waste	8–12 containers	15–30+ tons/day
Landfill remediation project	Sorting (high capacity) + Pyrolysis (multiple) + Metals	6–10 containers	10–25 tons/day

Each configuration is derived from the characterized feedstock inventory — not estimated from industry averages or assumed from facility design specifications. The domain is declared first. The configuration follows.

The ABRCE Framework Applied

The MAPD is not merely a modular processing system. It is an operational instance of the ABRCE invariant relational kernel applied to materials processing.

A — Domain Declaration: The local anthropogenic reservoir is characterized before any processing equipment is deployed. Feedstock composition, volume, contamination profile, and seasonal variation are explicitly declared. The processing domain is bounded.

B — Boundary Enforcement: Module 1 (Sorting) enforces the boundaries of what enters each processing stream. PVC is excluded from pyrolysis feedstock. Contaminated metals are routed to appropriate handling. Every downstream module receives input that is within its declared admissible domain. Formal condition: $T(D) \subseteq D$ — every processing transformation produces output that remains within the defined system.

R — Relational Structure: The relationships between modules are explicitly defined. Sorted output from Module 1 feeds specific processing modules based on material type.

Syngas from Module 2 routes to Module 5 for power generation. Metal output from Module 3 feeds local industrial demand or regional markets. These relationships constitute the depot's operational topology — and they are declared, not emergent.

C — Consequence (Invariants): Given the declared domain and relational structure, operational properties are derivable before deployment. Throughput capacity, energy balance (including syngas self-fueling contribution), output product mix, and break-even volume are structural consequences of the declared configuration — not empirical discoveries made after commissioning.

E — Execution (Enforcement): Runtime monitoring enforces domain closure during operation. If feedstock composition drifts from the declared domain — contamination levels rise, material mix shifts seasonally — the system produces an explicit signal rather than degrading silently. Operator adjustments are made against a known domain, not against unexplained performance changes.

Economics

The Energy Arbitrage at Depot Scale

The energy advantage of anthropogenic feedstock over geological extraction holds at any processing scale. It is a thermodynamic property of the material, not a property of facility size. What changes at small scale is the economic viability envelope — and the MAPD architecture is designed to operate within that envelope by matching processing capacity to actual local conditions.

Revenue Streams

Each MAPD generates revenue from multiple output products:

Pyrolysis oil: Current market price $600–900 per ton. Energy content of 42–50 MJ/kg — comparable to commercial diesel. Usable as industrial fuel, refinery feedstock, or chemical recycling input.

Syngas: Market value $200–300 per ton. Can generate up to 800 kWh of electricity per ton of waste processed. Can be consumed internally for power generation, reducing operating costs and potentially achieving energy self-sufficiency.

Carbon black: Market value $200–400 per ton. Industrial applications in rubber, plastics, and coatings.

Metal ingots: Aluminum, copper, and steel at secondary market prices — typically 70–90% of primary metal prices with substantially lower production costs.

E-waste concentrates: Precious metals and rare earths at commodity prices. Highest value per ton of any output stream.

Cost Structure

Capital expenditure — entry-level MAPD (1–3 tons/day):

The smallest commercially viable MAPD deployment — sorting module plus pyrolysis module plus shared utilities — represents a total capital investment of approximately $115,000–$300,000:

Module	Estimated Cost
Sorting/characterization (scaled NIR optical sorting, magnetic and eddy current separation)	$50,000–$150,000
Plastics pyrolysis (1 TPD containerized continuous unit)	$45,000–$100,000
Utilities and output handling	$20,000–$50,000
Entry-level MAPD total	**$115,000–$300,000**

Adding a metals reprocessing module (induction furnace, 50–250 kg batch capacity) adds $15,000–$50,000. Adding an e-waste recovery module for urban deployments adds $30,000–$80,000 depending on configuration.

For context: a skid-mounted pilot pyrolysis reactor for feedstock testing and proof-of-concept starts as low as $15,000–$21,000. A mid-scale MAPD deployment (5–10 TPD, sorting + multiple pyrolysis units + metals) falls in the $300,000–$800,000 range. These figures are based on 2024–2026 market data from commercial suppliers.

By comparison, centralized pyrolysis facilities at 50+ TPD scale require $5–$25 million in capital investment. The MAPD entry point is 1–2 orders of magnitude lower — deployable by small operators, municipal waste authorities, or regional investors without the capital concentration risk of centralized facilities.

Feedstock cost: $30–200 per ton for waste plastic, or negative cost where tipping fees are available. Scrap metal feedstock priced at local market rates.

Operating cost: Low labor requirements — containerized systems are designed for high automation with PLC control systems. Energy costs offset or eliminated by syngas self-fueling (syngas generates up to 800 kWh of electricity per ton of waste processed). Maintenance modular — individual modules can be serviced or replaced without shutting down the depot.

Logistics cost: Minimized by design. The depot is deployed at or near the feedstock source, eliminating the long-haul transportation that erodes centralized facility margins.

Break-Even

At the entry-level MAPD scale (1 TPD pyrolysis), daily output from plastic processing alone includes approximately 600–850 kg of pyrolysis oil (at $600–$900/ton), plus syngas (consumed internally or valued at $200–$300/ton) and carbon black ($200–$400/ton). Against a capital investment of $115,000–$300,000 and feedstock costs of $30–200 per ton (or negative with tipping fees), payback periods of 12–24 months are achievable under favorable feedstock conditions.

Containerized pyrolysis units are commercially viable at scales as low as 1 ton per day at current crude oil prices above $65 per barrel. Tipping fees, where available, improve the break-even point further. Metal reprocessing via induction furnace is profitable at even lower throughputs due to the extreme energy differential (5% for aluminum, 10–15% for copper versus primary production).

The MAPD architecture shifts the break-even calculation from "how much feedstock can we aggregate?" to "what does the local reservoir actually contain?" — and the answer, in most locations with any significant waste stream, is "enough."

Scaling Model

The MAPD scales by multiplication, not expansion.

A single depot proves the configuration against a local feedstock inventory. Once validated, the identical module configuration deploys to the next site with equivalent feedstock characteristics. Variations in feedstock composition across sites are addressed by varying the module configuration — not by redesigning the facility.

This produces a network topology: multiple bounded processing nodes, each operating within its declared domain, connected by output product logistics to regional markets. The network grows by adding nodes. Each node is independently viable. The failure of any single node does not compromise the network.

This is the inverse of the centralized model's risk profile. Centralized facilities concentrate capital and feedstock dependency. The MAPD network distributes both.

Growth Trajectory

Phase 1 — Proof of Configuration: Single deployment at a characterized site. Validate module performance, throughput, energy balance, and output quality against declared

domain specifications.

Phase 2 — Regional Replication: Deploy 3–5 depots across varied feedstock profiles within a region. Demonstrate configuration flexibility and output consistency. Establish regional output product logistics.

Phase 3 — Network Scale: Expand to 10+ depots. Output volumes reach levels that support direct relationships with refineries, foundries, and industrial buyers. Network effects emerge: shared maintenance, standardized compliance, bulk procurement of module components.

Phase 4 — Landfill Remediation: Deploy to legacy landfill sites for systematic extraction and processing of accumulated anthropogenic feedstock. This unlocks the largest untapped inventory — billions of tonnes of material currently classified as waste on no balance sheet.

Why This Matters Now

The containerized pyrolysis market was valued at $1.46 billion in 2024 and is projected to reach $3.04 billion by 2033, growing at 8.7% annually. Over 430 mobile pyrolysis units have been commissioned globally. Modular plant designs saw 47% adoption growth in recent years. The equipment is commercially available, the technology is proven, and the market is scaling.

What is missing is the architectural layer that ensures each deployment operates within a bounded, declared domain — that feedstock characterization precedes processing deployment, that module configurations match actual inventories, and that runtime operations enforce domain closure rather than discovering operational limits empirically.

That architectural layer is what Metatron Dynamics provides.

Investment Thesis

The MAPD represents an investment in distributed infrastructure positioned on the correct side of a structural energy-cost gradient.

The gradient is permanent. Anthropogenic feedstock retains the energy investment of primary processing. Reprocessing requires a fraction of that original energy. This advantage is thermodynamic — it does not depend on regulation, subsidies, or consumer preference.

The gradient is widening. Geological extraction targets increasingly depleted, remote, and energy-intensive deposits. The anthropogenic reservoir grows with every product manufactured. The cost curves are converging, and for aluminum and steel, they have already crossed.

The architecture is capital-efficient. Containerized modules deploy at a fraction of centralized facility cost, with deployment timelines measured in days. Capital risk is distributed across multiple independent sites rather than concentrated in single facilities.

The architecture is domain-bounded. Every deployment operates against a declared feedstock domain with enforced processing boundaries. This is not a theoretical distinction — it is the operational difference between discovering what works through commissioning failures and knowing what works through structural analysis before the first module arrives.

The network scales by multiplication. Each proven configuration replicates to equivalent sites. Growth is linear in capital and logarithmic in risk.

Conclusion

The anthropogenic reservoir is the largest, lowest-energy, lowest-risk feedstock source available to the global economy. The technology to process it at containerized scale exists and is commercially proven. The missing layer is the architectural discipline that ensures each deployment is bounded, configured to its actual domain, and enforced during operation.

The Modular Anthropogenic Processing Depot applies that discipline — deploying scale-appropriate reprocessing to where the feedstock already is, configured to what is actually there, and governed by the structural guarantees that prevent every deployment from becoming an empirical experiment.

Unbounded systems can be made to work. Bounded systems can be made to last. The difference is whether validity is assumed — or enforced.

Metatron Dynamics Relational analysis applied to structural reality.
relationalrelativity.dev | GitHub: Relational-Relativity-Corporation